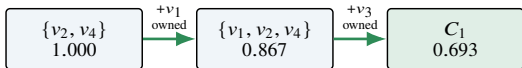


**Feasible parent.** For feasible  $D$ , delete the vertex  $d(D)$  with minimum total similarity. Then  $\text{avgSim}(D \setminus \{d(D)\}) \geq \text{avgSim}(D)$ .

**Unique ownership.** Traverse  $C \rightarrow D = C \cup \{x\}$  only if  $x = d(D)$ . Thus each state has one parent and needs no visited table.

**Owned path to  $C_1$**



**Two paths to  $C_2$**

**Traverse**

$\{v_5, v_6\} + v_3 \rightarrow C_2$   
 $v_3 = d(C_2)$

**Stop duplicate path**

$\{v_3, v_5\} + v_6 \rightarrow C_2$   
 $v_6 \neq d(C_2) = v_3$

Reverse search reaches each feasible state once;  
the final containment filter returns  $C_1$  and  $C_2$ .